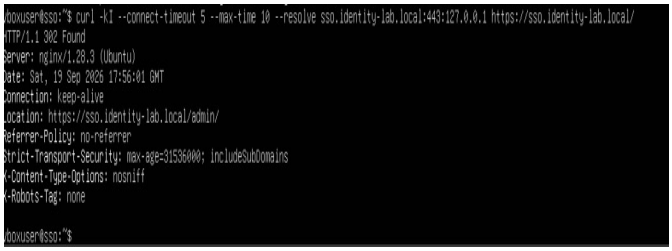
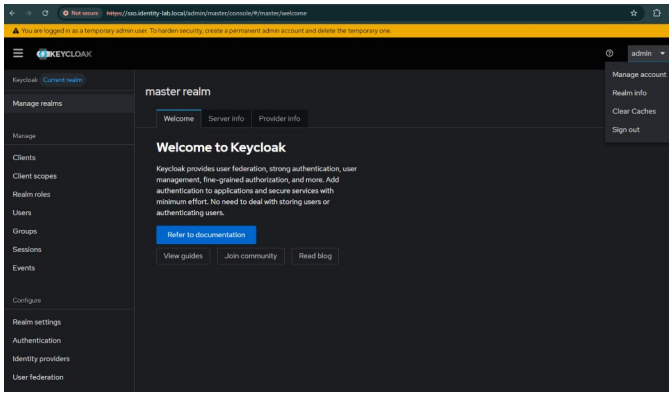
00 — HTTPS, NGINX, and Keycloak Connectivity Check

I tested the local HTTPS endpoint using curl with the hostname sso.identity-lab.local. NGINX returned an HTTP 302 response redirecting to the Keycloak administration page, confirming that the HTTPS reverse proxy could reach Keycloak. The test used the -k option because the certificate was self-signed.



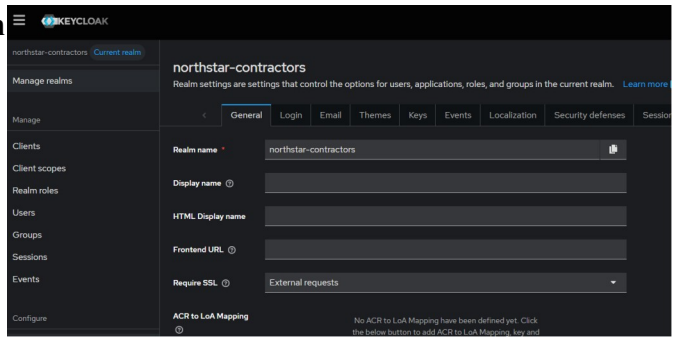
01 — Keycloak Administration Console

I signed in to the Keycloak Administration Console using the administrator account. The console opened in the master realm, providing access to realm management and identity provider configuration.



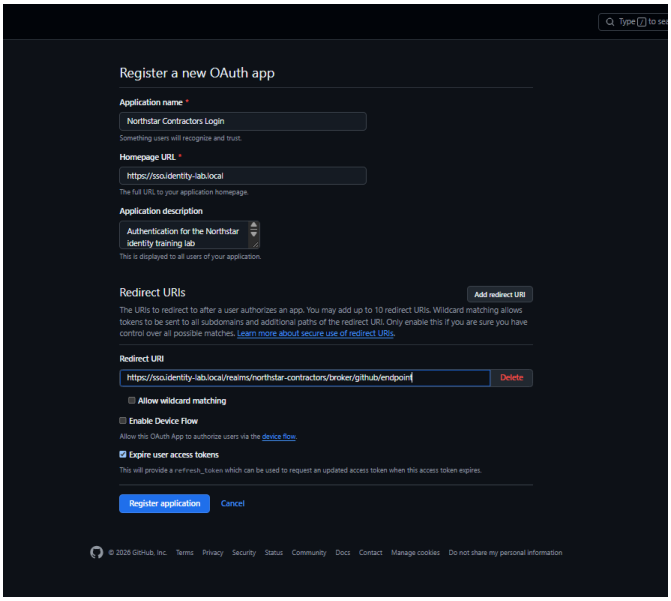
02 — Creating the Northstar Contractors Realm

I created a dedicated realm named northstar-contractors. This realm separates the laboratory's external authentication configuration and user accounts from the master realm.



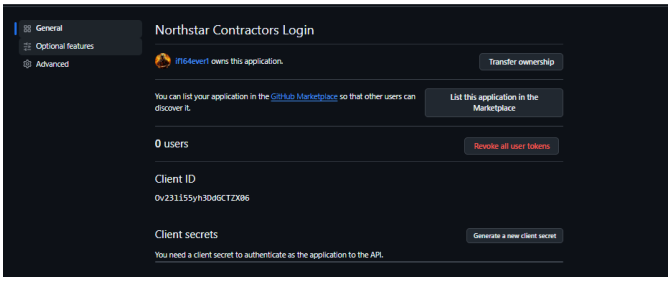
03 — GitHub OAuth Application Registration

I completed the GitHub OAuth application registration form using the name Northstar Contractors Login and the homepage URL https://sso.identity-lab.local. I entered https://sso.identity-lab.local/realms/northstar-contractors/broker/github/endpoint as the redirect URI to return users to Keycloak after GitHub authorization



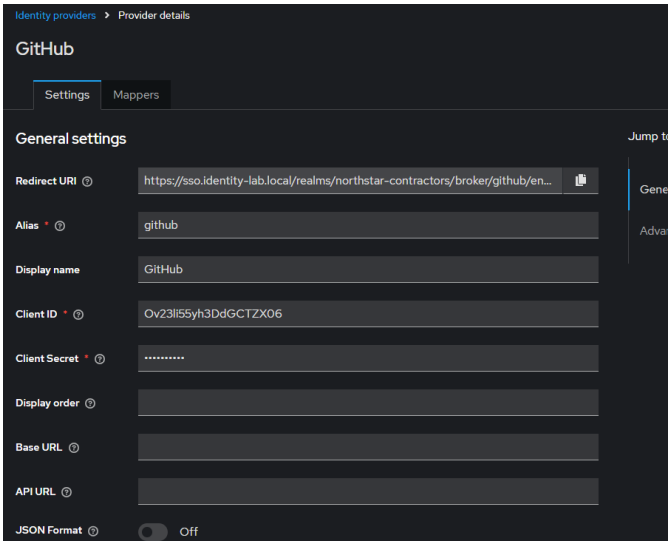
04 — GitHub OAuth Application Client ID

I registered the OAuth application and obtained its Client ID. I then generated a Client Secret for configuring the connection in Keycloak; the secret is not exposed in the screenshot.



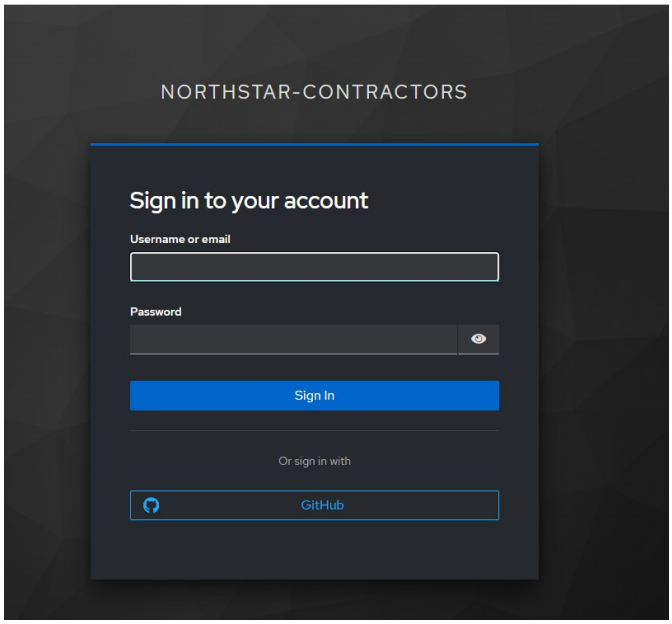
05 — Configuring the GitHub Identity Provider

I added GitHub as an identity provider in the northstar-contractors realm using the alias github. I entered the OAuth application's Client ID and Client Secret and verified that the provider's redirect URI matched the address registered in GitHub.



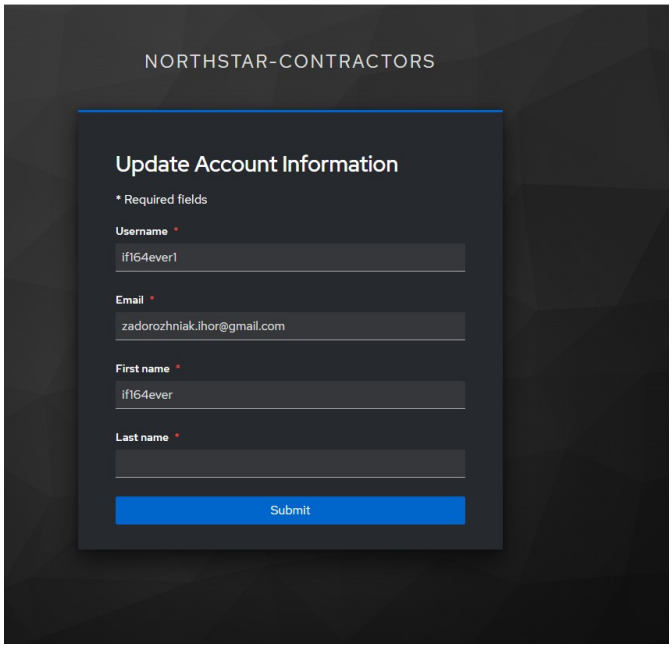
06 — Starting Login with GitHub

I opened the Account Console for the northstar-contractors realm in a private browser window. I selected the GitHub sign-in option on the Keycloak login page to begin authentication through the external provider.



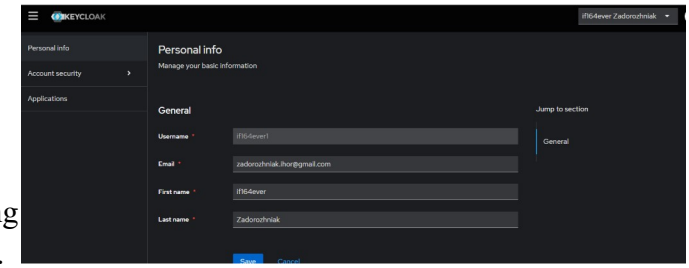
07 — GitHub Application Authorization

GitHub displayed an authorization request for the Northstar Contractors Login application. I approved the requested access, allowing the authentication process to continue and the browser to return to Keycloak.



08 — Successful Login through GitHub

I successfully accessed the Keycloak Account Console after authenticating through GitHub. The Personal info page displayed the username if164ever1 and the user's profile details, confirming successful login to the northstar-contractors realm.



09 — Verifying the GitHub Identity Link

I opened the user's **Identity provider links** tab in the Keycloak Administration Console. GitHub appeared under **Linked identity providers** with the username if164ever1, confirming that the Keycloak user account was linked to the external GitHub identity.

